```
!pip install gradio transformers accelerate torch
```

```
Requirement already satisfied: safehttpx<0.2.0,>=0.1.6 in /usr/local/lib/python3.12/dist-packages (from gradio) (0.1.7)
Requirement already satisfied: semantic-version~=2.0 in /usr/local/lib/python3.12/dist-packages (from gradio) (2.10.0)
Requirement already satisfied: starlette<1.0,>=0.40.0 in /usr/local/lib/python3.12/dist-packages (from gradio) (0.52.1)
Requirement already satisfied: tomlkit<0.14.0,>=0.12.0 in /usr/local/lib/python3.12/dist-packages (from gradio) (0.13.3)
Requirement already satisfied: typer<1.0,>=0.12 in /usr/local/lib/python3.12/dist-packages (from gradio) (0.24.1)
Requirement already satisfied: typing-extensions~=4.0 in /usr/local/lib/python3.12/dist-packages (from gradio) (4.15.0)
Requirement already satisfied: uvicorn>=0.14.0 in /usr/local/lib/python3.12/dist-packages (from gradio) (0.41.0)
Requirement already satisfied: fsspec in /usr/local/lib/python3.12/dist-packages (from gradio-client==1.14.0->gradio) (202
Requirement already satisfied: websockets<16.0,>=13.0 in /usr/local/lib/python3.12/dist-packages (from gradio-client==1.14
Requirement already satisfied: filelock in /usr/local/lib/python3.12/dist-packages (from transformers) (3.24.3)
Requirement already satisfied: regex!=2019.12.17 in /usr/local/lib/python3.12/dist-packages (from transformers) (2025.11.3
Requirement already satisfied: tokenizers<=0.23.0,>=0.22.0 in /usr/local/lib/python3.12/dist-packages (from transformers)
Requirement already satisfied: typer-slim in /usr/local/lib/python3.12/dist-packages (from transformers) (0.24.0)
Requirement already satisfied: safetensors>=0.4.3 in /usr/local/lib/python3.12/dist-packages (from transformers) (0.7.0)
Requirement already satisfied: tqdm>=4.27 in /usr/local/lib/python3.12/dist-packages (from transformers) (4.67.3)
Requirement already satisfied: psutil in /usr/local/lib/python3.12/dist-packages (from accelerate) (5.9.5)
Requirement already satisfied: setuptools in /usr/local/lib/python3.12/dist-packages (from torch) (75.2.0)
Requirement already satisfied: sympy>=1.13.3 in /usr/local/lib/python3.12/dist-packages (from torch) (1.14.0)
Requirement already satisfied: networkx>=2.5.1 in /usr/local/lib/python3.12/dist-packages (from torch) (3.6.1)
Requirement already satisfied: cuda-bindings==12.9.4 in /usr/local/lib/python3.12/dist-packages (from torch) (12.9.4)
Requirement already satisfied: nvidia-cuda-nvrtc-cu12==12.8.93 in /usr/local/lib/python3.12/dist-packages (from torch) (12
Requirement already satisfied: nvidia-cuda-runtime-cu12==12.8.90 in /usr/local/lib/python3.12/dist-packages (from torch) (
Requirement already satisfied: nvidia-cuda-cupti-cu12==12.8.90 in /usr/local/lib/python3.12/dist-packages (from torch) (12
Requirement already satisfied: nvidia-cudnn-cu12==9.10.2.21 in /usr/local/lib/python3.12/dist-packages (from torch) (9.10.
Requirement already satisfied: nvidia-cublas-cu12==12.8.4.1 in /usr/local/lib/python3.12/dist-packages (from torch) (12.8.
Requirement already satisfied: nvidia-cufft-cu12==11.3.3.83 in /usr/local/lib/python3.12/dist-packages (from torch) (11.3.
Requirement already satisfied: nvidia-curand-cu12==10.3.9.90 in /usr/local/lib/python3.12/dist-packages (from torch) (10.3
Requirement already satisfied: nvidia-cusolver-cu12==11.7.3.90 in /usr/local/lib/python3.12/dist-packages (from torch) (11
Requirement already satisfied: nvidia-cusparse-cu12==12.5.8.93 in /usr/local/lib/python3.12/dist-packages (from torch) (12
Requirement already satisfied: nvidia-cusparselt-cu12==0.7.1 in /usr/local/lib/python3.12/dist-packages (from torch) (0.7.
Requirement already satisfied: nvidia-nccl-cu12==2.27.5 in /usr/local/lib/python3.12/dist-packages (from torch) (2.27.5)
Requirement already satisfied: nvidia-nvshmem-cu12==3.4.5 in /usr/local/lib/python3.12/dist-packages (from torch) (3.4.5)
Requirement already satisfied: nvidia-nvtx-cu12==12.8.90 in /usr/local/lib/python3.12/dist-packages (from torch) (12.8.90)
Requirement already satisfied: nvidia-nvjitlink-cu12==12.8.93 in /usr/local/lib/python3.12/dist-packages (from torch) (12.
Requirement already satisfied: nvidia-cufile-cu12==1.13.1.3 in /usr/local/lib/python3.12/dist-packages (from torch) (1.13.
Requirement already satisfied: triton==3.6.0 in /usr/local/lib/python3.12/dist-packages (from torch) (3.6.0)
Requirement already satisfied: cuda-pathfinder~=1.1 in /usr/local/lib/python3.12/dist-packages (from cuda-bindings==12.9.4
Requirement already satisfied: idna>=2.8 in /usr/local/lib/python3.12/dist-packages (from anyio<5.0,>=3.0->gradio) (3.11)
Requirement already satisfied: typing-inspection>=0.4.2 in /usr/local/lib/python3.12/dist-packages (from fastapi<1.0,>=0.1
Requirement already satisfied: annotated-doc>=0.0.2 in /usr/local/lib/python3.12/dist-packages (from fastapi<1.0,>=0.115.2
Requirement already satisfied: certifi in /usr/local/lib/python3.12/dist-packages (from httpx<1.0,>=0.24.1->gradio) (2026.
Requirement already satisfied: httpcore==1.* in /usr/local/lib/python3.12/dist-packages (from httpx<1.0,>=0.24.1->gradio)
Requirement already satisfied: h11>=0.16 in /usr/local/lib/python3.12/dist-packages (from httpcore==1.*->httpx<1.0,>=0.24.
Requirement already satisfied: hf-xet<2.0.0,>=1.2.0 in /usr/local/lib/python3.12/dist-packages (from huggingface-hub<2.0,>
Requirement already satisfied: python-dateutil>=2.8.2 in /usr/local/lib/python3.12/dist-packages (from pandas<3.0,>=1.0->g
Requirement already satisfied: pytz>=2020.1 in /usr/local/lib/python3.12/dist-packages (from pandas<3.0,>=1.0->gradio) (20
Requirement already satisfied: tzdata>=2022.7 in /usr/local/lib/python3.12/dist-packages (from pandas<3.0,>=1.0->gradio) (
Requirement already satisfied: annotated-types>=0.6.0 in /usr/local/lib/python3.12/dist-packages (from pydantic<=2.12.3,>=
Requirement already satisfied: pydantic-core==2.41.4 in /usr/local/lib/python3.12/dist-packages (from pydantic<=2.12.3,>=2
Requirement already satisfied: mpmath<1.4,>=1.1.0 in /usr/local/lib/python3.12/dist-packages (from sympy>=1.13.3->torch) (
Requirement already satisfied: click>=8.2.1 in /usr/local/lib/python3.12/dist-packages (from typer<1.0,>=0.12->gradio) (8.
Requirement already satisfied: shellingham>=1.3.0 in /usr/local/lib/python3.12/dist-packages (from typer<1.0,>=0.12->gradi
Requirement already satisfied: rich>=12.3.0 in /usr/local/lib/python3.12/dist-packages (from typer<1.0,>=0.12->gradio) (13
Requirement already satisfied: six>=1.5 in /usr/local/lib/python3.12/dist-packages (from python-dateutil>=2.8.2->pandas<3.
Requirement already satisfied: markdown-it-py>=2.2.0 in /usr/local/lib/python3.12/dist-packages (from rich>=12.3.0->typer<
Requirement already satisfied: pygments<3.0.0,>=2.13.0 in /usr/local/lib/python3.12/dist-packages (from rich>=12.3.0->type
Requirement already satisfied: mdurl~=0.1 in /usr/local/lib/python3.12/dist-packages (from markdown-it-py>=2.2.0->rich>=12
```

```
!pip install sentencepiece
```

```
Requirement already satisfied: sentencepiece in /usr/local/lib/python3.12/dist-packages (0.2.1)
```

```
!pip install gradio transformers accelerate torch
```

```
Requirement already satisfied: cuda-bindings==12.9.4 in /usr/local/lib/python3.12/dist-packages (from torch) (12.9.4)
Requirement already satisfied: nvidia-cuda-nvrtc-cu12==12.8.93 in /usr/local/lib/python3.12/dist-packages (from torch) (12
Requirement already satisfied: nvidia-cuda-runtime-cu12==12.8.90 in /usr/local/lib/python3.12/dist-packages (from torch) (
Requirement already satisfied: nvidia-cuda-cupti-cu12==12.8.90 in /usr/local/lib/python3.12/dist-packages (from torch) (12
Requirement already satisfied: nvidia-cudnn-cu12==9.10.2.21 in /usr/local/lib/python3.12/dist-packages (from torch) (9.10.
Requirement already satisfied: nvidia-cublas-cu12==12.8.4.1 in /usr/local/lib/python3.12/dist-packages (from torch) (12.8.
Requirement already satisfied: nvidia-cufft-cu12==11.3.3.83 in /usr/local/lib/python3.12/dist-packages (from torch) (11.3.
Requirement already satisfied: nvidia-curand-cu12==10.3.9.90 in /usr/local/lib/python3.12/dist-packages (from torch) (10.3
Requirement already satisfied: nvidia-cusolver-cu12==11.7.3.90 in /usr/local/lib/python3.12/dist-packages (from torch) (11
Requirement already satisfied: nvidia-cusparse-cu12==12.5.8.93 in /usr/local/lib/python3.12/dist-packages (from torch) (12
Requirement already satisfied: nvidia-cusparselt-cu12==0.7.1 in /usr/local/lib/python3.12/dist-packages (from torch) (0.7.
Requirement already satisfied: nvidia-nccl-cu12==2.27.5 in /usr/local/lib/python3.12/dist-packages (from torch) (2.27.5)
Requirement already satisfied: nvidia-nvshmem-cu12==3.4.5 in /usr/local/lib/python3.12/dist-packages (from torch) (3.4.5)
Requirement already satisfied: nvidia-nvtx-cu12==12.8.90 in /usr/local/lib/python3.12/dist-packages (from torch) (12.8.90)
Requirement already satisfied: nvidia-nvjitlink-cu12==12.8.93 in /usr/local/lib/python3.12/dist-packages (from torch) (12.
Requirement already satisfied: nvidia-cufile-cu12==1.13.1.3 in /usr/local/lib/python3.12/dist-packages (from torch) (1.13.
Requirement already satisfied: triton==3.6.0 in /usr/local/lib/python3.12/dist-packages (from torch) (3.6.0)
Requirement already satisfied: cuda-pathfinder~=1.1 in /usr/local/lib/python3.12/dist-packages (from cuda-bindings==12.9.4
Requirement already satisfied: idna>=2.8 in /usr/local/lib/python3.12/dist-packages (from anyio<5.0,>=3.0->gradio) (3.11)
Requirement already satisfied: typing-inspection>=0.4.2 in /usr/local/lib/python3.12/dist-packages (from fastapi<1.0,>=0.1
Requirement already satisfied: annotated-doc>=0.0.2 in /usr/local/lib/python3.12/dist-packages (from fastapi<1.0,>=0.115.2
Requirement already satisfied: certifi in /usr/local/lib/python3.12/dist-packages (from httpx<1.0,>=0.24.1->gradio) (2026.
Requirement already satisfied: httpcore==1.* in /usr/local/lib/python3.12/dist-packages (from httpx<1.0,>=0.24.1->gradio)
Requirement already satisfied: h11>=0.16 in /usr/local/lib/python3.12/dist-packages (from httpcore==1.*->httpx<1.0,>=0.24.
Requirement already satisfied: hf-xet<2.0.0,>=1.2.0 in /usr/local/lib/python3.12/dist-packages (from huggingface-hub<2.0,>
Requirement already satisfied: python-dateutil>=2.8.2 in /usr/local/lib/python3.12/dist-packages (from pandas<3.0,>=1.0->g
Requirement already satisfied: pytz>=2020.1 in /usr/local/lib/python3.12/dist-packages (from pandas<3.0,>=1.0->gradio) (20
Requirement already satisfied: tzdata>=2022.7 in /usr/local/lib/python3.12/dist-packages (from pandas<3.0,>=1.0->gradio) (
Requirement already satisfied: annotated-types>=0.6.0 in /usr/local/lib/python3.12/dist-packages (from pydantic<=2.12.3,>=
Requirement already satisfied: pydantic-core==2.41.4 in /usr/local/lib/python3.12/dist-packages (from pydantic<=2.12.3,>=2
Requirement already satisfied: mpmath<1.4,>=1.1.0 in /usr/local/lib/python3.12/dist-packages (from sympy>=1.13.3->torch) (
Requirement already satisfied: click>=8.2.1 in /usr/local/lib/python3.12/dist-packages (from typer<1.0,>=0.12->gradio) (8.
Requirement already satisfied: shellingham>=1.3.0 in /usr/local/lib/python3.12/dist-packages (from typer<1.0,>=0.12->gradi
Requirement already satisfied: rich>=12.3.0 in /usr/local/lib/python3.12/dist-packages (from typer<1.0,>=0.12->gradio) (13
Requirement already satisfied: six>=1.5 in /usr/local/lib/python3.12/dist-packages (from python-dateutil>=2.8.2->pandas<3.
Requirement already satisfied: markdown-it-py>=2.2.0 in /usr/local/lib/python3.12/dist-packages (from rich>=12.3.0->typer<
Requirement already satisfied: pygments<3.0.0,>=2.13.0 in /usr/local/lib/python3.12/dist-packages (from rich>=12.3.0->type
Requirement already satisfied: mdurl~=0.1 in /usr/local/lib/python3.12/dist-packages (from markdown-it-py>=2.2.0->rich>=12
```

```
model_id = "ibm-granite/granite-3.3-2b-instruct"
```

```
from transformers import AutoModelForCausalLM, AutoTokenizer
import torch

model_id = "ibm-granite/granite-3.3-2b-instruct"

tokenizer = AutoTokenizer.from_pretrained(model_id)
model = AutoModelForCausalLM.from_pretrained(
    model_id,
    device_map="auto",
    torch_dtype="auto"
)

print(f"Model {model_id} loaded successfully.")
```

```
Loading weights: 100%                                    362/362 [00:02<00:00, 143.27it/s, Materializing param=model.norm.weight]
Model ibm-granite/granite-3.3-2b-instruct loaded successfully.
```

```
from transformers import AutoTokenizer, AutoModelForCausalLM
import torch

model_name = "ibm-granite/granite-3.3-2b-instruct"

tokenizer = AutoTokenizer.from_pretrained(model_name)
model = AutoModelForCausalLM.from_pretrained(
    model_name,
    torch_dtype=torch.float16,
    device_map="auto"
)

def generate_text(prompt):
    inputs = tokenizer(prompt, return_tensors="pt").to("cuda")
    outputs = model.generate(**inputs, max_new_tokens=300)
    return tokenizer.decode(outputs[0], skip_special_tokens=True)

print("Model Loaded Successfully ✅")
```

```
Loading weights: 100%                                    362/362 [00:05<00:00, 52.66it/s, Materializing param=model.norm.weight]
WARNING:accelerate.big_modeling:Some parameters are on the meta device because they were offloaded to the cpu.
Model Loaded Successfully ✅
```

```
class ColorPaletteGenerator:
    def __init__(self):
        self.palettes = {
            "modern": {
                "primary": "#4F46E5",
                "secondary": "#6366F1",
                "accent": "#22D3EE",
                "background": "#0F172A",
                "surface": "#1E293B",
                "text_primary": "#F8FAFC",
                "text_secondary": "#CBD5E1",
                "border": "#334155"
            },
            "vibrant": {
                "primary": "#FF6B6B",
                "secondary": "#FFD93D",
                "accent": "#6BCB77",
                "background": "#1A1A2E",
                "surface": "#16213E",
                "text_primary": "#FFFFFF",
                "text_secondary": "#EAEAEA",
                "border": "#0F3460"
            },
            "ocean": {
                "primary": "#0077B6",
                "secondary": "#0096C7",
                "accent": "#00B4D8",
                "background": "#03045E",
                "surface": "#023E8A",
                "text_primary": "#CAF0F8",
                "text_secondary": "#ADE8F4",
                "border": "#0077B6"
            }
        }

    def get_palette(self, style):
        return self.palettes.get(style, self.palettes["modern"])
```

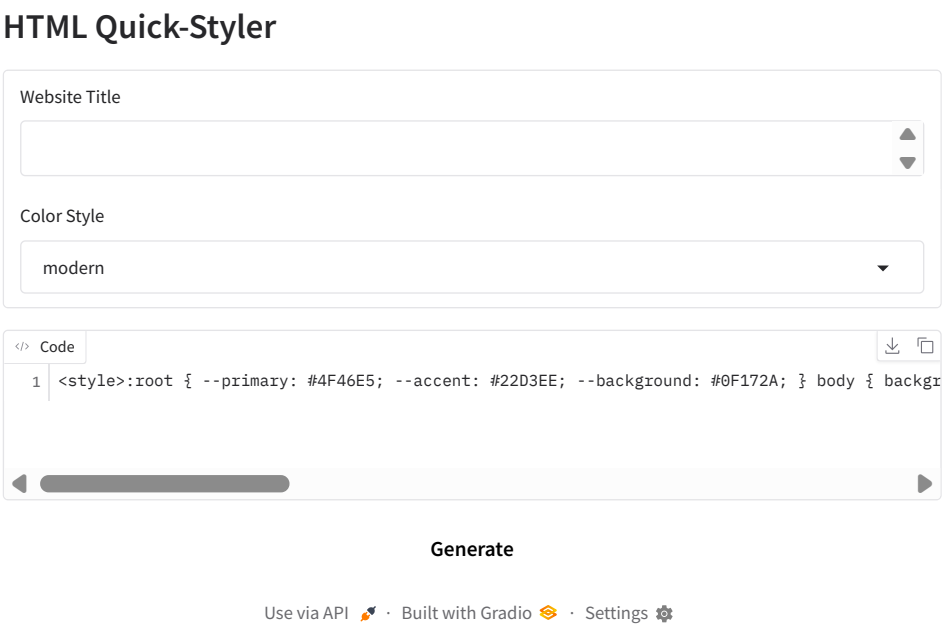
```
class HTMLPageBuilder:

    def generate_meta_tags(self, title):
        return f"""
        <meta charset="UTF-8">
        <meta name="viewport" content="width=device-width, initial-scale=1.0">
        <title>{title}</title>
        <meta property="og:title" content="{title}">
        <meta name="twitter:card" content="summary_large_image">
        """

    def build_hero_section(self, title):
        return f"""
        <section class="hero">
            <h1>{title}</h1>
            <p>Welcome to our modern AI generated website</p>
            <button>Get Started</button>
        </section>
        """

    def build_footer(self):
        return """
        <footer>
            <p>© 2026 HTML Quick-Styler. All Rights Reserved.</p>
        </footer>
        """

    def build_page(self, title):
        meta = self.generate_meta_tags(title)
        hero = self.build_hero_section(title)
        footer = self.build_footer()

        return f"""
        <!DOCTYPE html>
        <html>
        <head>
        {meta}
        </head>
        <body>
        {hero}
        {footer}
        </body>
        </html>
        """
```

```
def generate_css(colors):
    return f"""
    :root {{
        --primary: {colors['primary']};
        --secondary: {colors['secondary']};
        --accent: {colors['accent']};
        --background: {colors['background']};
        --surface: {colors['surface']};
        --text-primary: {colors['text_primary']};
        --text-secondary: {colors['text_secondary']};
    }}

    body {{
        margin: 0;
        font-family: Arial, sans-serif;
        background: var(--background);
        color: var(--text-primary);
    }}

    .hero {{
        text-align: center;
        padding: 100px 20px;
        background: linear-gradient(135deg, var(--primary), var(--secondary));
    }}

    button {{
        padding: 12px 24px;
        background: var(--accent);
        border: none;
        border-radius: 5px;
        cursor: pointer;
        transition: 0.3s;
    }}

    button:hover {{
        opacity: 0.8;
    }}
```

```
    footer {{
        text-align: center;
        padding: 20px;
        background: var(--surface);
    }}
    """
```

```
import gradio as gr

# --- Redefining dependencies to ensure they exist in the current session ---
class ColorPaletteGenerator:
    def __init__(self):
        self.palettes = {
            "modern": {"primary": "#4F46E5", "secondary": "#6366F1", "accent": "#22D3EE", "background": "#0F172A", "surface
            "vibrant": {"primary": "#FF6B6B", "secondary": "#FFD93D", "accent": "#6BCB77", "background": "#1A1A2E", "surfac
            "ocean": {"primary": "#0077B6", "secondary": "#0096C7", "accent": "#00B4D8", "background": "#03045E", "surface'
        }
    def get_palette(self, style):
        return self.palettes.get(style, self.palettes["modern"])

class HTMLPageBuilder:
    def build_page(self, title):
        return f"<!DOCTYPE html><html><head><title>{title}</title></head><body><section class='hero'><h1>{title}</h1><butto

def generate_css(colors):
    return f":root {{ --primary: {colors['primary']}; --accent: {colors['accent']}; --background: {colors['background']}; 

# --- Main App Logic ---
palette_generator = ColorPaletteGenerator()
page_builder = HTMLPageBuilder()

def generate_website(title, style):
    colors = palette_generator.get_palette(style)
    html = page_builder.build_page(title)
    css = generate_css(colors)
    return f"<style>{css}</style>" + html

with gr.Blocks() as demo:
    gr.Markdown("# HTML Quick-Styler")
    title = gr.Textbox(label="Website Title")
    style = gr.Dropdown(["modern", "vibrant", "ocean"], label="Color Style")
    output = gr.Code(language="html")
    btn = gr.Button("Generate")
    btn.click(generate_website, inputs=[title, style], outputs=output)

demo.launch(share=True)
```

```
Colab notebook detected. To show errors in colab notebook, set debug=True in launch()
* Running on public URL: https://f6101e28662e05fb23.gradio.live

This share link expires in 1 week. For free permanent hosting and GPU upgrades, run `gradio deploy` from the terminal in the
```



```
demo.launch(share=True)
```

```
Rerunning server... use `close()` to stop if you need to change `launch()` parameters.
----
Colab notebook detected. To show errors in colab notebook, set debug=True in launch()
* Running on public URL: https://f6101e28662e05fb23.gradio.live

This share link expires in 1 week. For free permanent hosting and GPU upgrades, run `gradio deploy` from the terminal in the
```

HTML Quick-Styler

Website Title

Modern Portfolio

Color Style

modern

Code

```
1 <style>:root { --primary: #4F46E5; --accent: #22D3EE; --background: #0F172A; } body { backgr
```

Generate

Use via API · Built with Gradio · Settings

Next steps: Deploy to Cloud Run

Gemini

```
import os
from google.colab import drive
drive.mount('/content/drive', force_remount=True)

# Check if drive is already mounted to avoid unnecessary prompts
if not os.path.exists('/content/drive'):
    try:
        drive.mount('/content/drive', force_remount=True)
```

Could not connect to the reCAPTCHA service. Please check your internet connection and reload to get a reCAPTCHA challenge.